UE24CS320A – Capstone Project Approval

Third Eye - A Closed-Loop Multimodal System for Early-to-Advanced Dementia Monitoring & Adaptive Care

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Outline

- Problem Statement
- Scope and Feasibility study
- Applications/Use cases
- Background work
- Expected Deliverables
- Capstone (Phase-I Phase-II, Phase-III) Project Timeline
- Any other information

Problem Statement

- Dementia progressively affects everyday functioning and routines, making **independent living** difficult. **Existing systems** mainly focus on **monitoring**, **activity recognition**, anomaly detection, or **caregiver alerts**, with **limited adaptation** to each individual's changing needs.

- A system is needed to distinguish **normal individual behaviour from genuine difficulty**, understand the person's **current context and state**, and determine **when and what** assistance is required, thereby supporting **greater independence** in daily living.



Detecting an abnormal event is not enough.
A **behavioural change** alone does not indicate a problem. Its meaning depends on the individual, context, and current state; therefore, identifying the change is only the first step—understanding it and determining the appropriate support is essential.

NEED

Can a system learn an individual's normal behaviour, identify meaningful changes, understand their context, and adapt assistance based on their response?

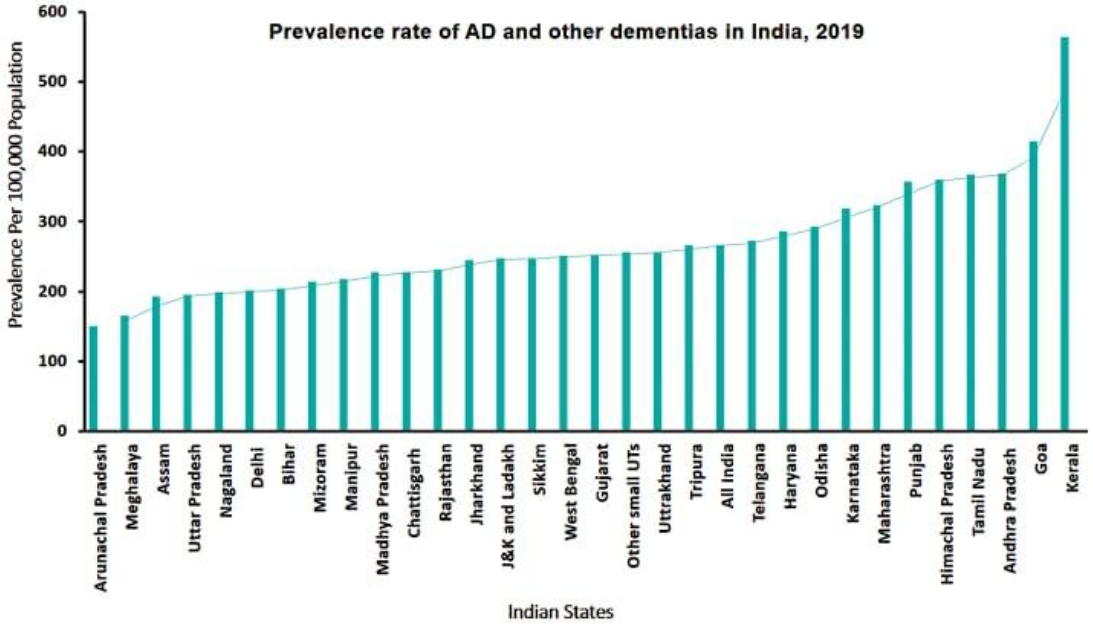
Motivation

Global

- 57 million people were living with dementia in 2021.
- Nearly 10 million new cases/year.
- Global dementia burden is projected to reach 78M by 2030 and 139M by 2050.

India

- An estimated 8.8 million Indians aged 60+ were living with dementia based on LASI data.
- Estimated prevalence among Indians aged 60+ is 7.4%
- Dementia in India estimates reported 5.3M in 2020, projected to exceed 14M by 2050.



Major Takeaway: Dementia is a growing global and Indian challenge, increasing the need for personalized support that helps people remain independent in daily living

References

https://www.who.int/news-room/fact-sheets/detail/dementia
https://alz-journals.onlinelibrary.wiley.com
https://ruralindiaonline.org/media/documents/Dementia_in_India_2020.pdf

Image Source

https://niscpr.res.in/includes/images/bulletin/bulletin-2023-03-15.pdf

Scope and Feasibility study

Scope

- **Personalized** dementia support across changing stages of cognitive decline.
- **Multimodal sensing**: ambient sensors, wearables, location and physiological signals.
- **Activity & routine understanding**: learn individual ADLs, routines and behavioural patterns.
- **Continuous learning**: learn from patient responses and update the personalized model.
- **Prototype**: A physical prototype that actively collects readings from patient, thereby facilitating reinforcement learning.

Feasibility

- **Data: CASAS** and **TIHM** datasets containing behaviour and movement tracking data, **Dem@Care** and **HAR** datasets for physiological (initial development)
- **AI/ML:** Activity recognition, anomaly/deviation detection and behavioural modelling are technically feasible using **existing ML methods**.
- **Hardware:** PIR, door/contact and wearable sensors are relatively accessible for prototyping.
- **Incremental development:** System can be developed module-by-module before full integration.
- **Key challenges:** Data sparsity, multimodal data integration, behavioural drift, sensor reliability and real-world validation.

One Line Conclusion: The project is technically feasible as a **phased prototype**, with large-scale real-world deployment and validation considered as future extensions.

References
https://zenodo.org/records/15708568 https://pubmed.ncbi.nlm.nih.gov/29723236/

Literature Study

Paper / Study	Problem	Approach / Method	Limitations	Relevance / Extension for Third Eye
Health Management and Pattern Analysis of Daily Living Activities using In-home Sensors and ML (2018)	Conventional assessment may miss subtle changes in daily routines and behaviour.	Uses in-home sensors + physiological data + ML to monitor routines and detect abnormalities.	Requires continuous sensing; mainly focuses on monitoring and alerts, with limited contextual understanding.	Third Eye extends this through multimodal sensing, contextual reasoning and personalized assistance.
Assessing Cognitive Decline through ADL Monitoring in an IoT Smart Home (2024)	Traditional tests may not capture functional changes in early cognitive decline.	Uses IoT sensors to monitor ADLs and analyses task duration/performance. Meal preparation achieved AUC ≈ 0.78–0.86 for group discrimination.	Simple tasks may not reveal subtle impairment; limited real-time intervention and longitudinal adaptation.	Supports Third Eye's use of complex daily activities as behavioural indicators and Digital Twin-based monitoring.
Personalized Behavioral Abnormality Detection in Smart Homes (2025)	Generic activity models can generate false alarms because routines differ between individuals.	Uses personalized HAR + LSTM/BiLSTM + ontology/rule-based reasoning to detect abnormal behaviour.	Requires personalization; mainly detects deviations rather than providing adaptive assistance.	Third Eye builds a personalized behavioural baseline and combines it with contextual reasoning and assistance.
Routine-Deviation Detection in Smart-Home Sensor Networks using GRU (2024)	Fixed anomaly thresholds can produce false alarms when normal routines vary.	Uses GRU prediction to learn expected individual routines and identify deviations.	Limited context; deviations are not always cognitive abnormalities and require further interpretation.	Third Eye combines routine prediction + multimodal context + cognitive-state modelling.
Human-Centered Ambient and Wearable Sensing for Automated Dementia Care: A Scoping Review (2026)	Existing dementia monitoring approaches face challenges in personalization, usability and continuous care.	Reviews ambient/wearable sensing, AI, privacy and human-centred approaches.	Highlights limited long-term validation, personalization and patient-centred intervention.	Establishes the need for multimodal, personalized, privacy-aware and longitudinal care systems such as Third Eye.

Background

Why this Problem??

- Dementia progressively affects **everyday functioning**, making familiar activities **increasingly difficult** over time.
- Tasks that were once completed independently may gradually require caregiver assistance or supervision.Current assistive systems often emphasize monitoring, activity recognition, anomaly detection, or **caregiver alerts**.
- A **behavioural change** does not necessarily indicate a problem, its meaning depends on the individual, context, routine, and current state.
- This creates a gap between detecting that something changed and understanding what the change means and providing appropriate assistance.
- This motivated us to explore a system that can **learn the individual**, **adapt to changing needs**, and **support independence in daily living**.

Domain Context

- **Dementia & Cognitive Decline**: Progressive changes in memory, cognition, behaviour and daily functioning.
- **Activities of Daily Living (ADLs)**: Everyday tasks such as eating, cooking, dressing, medication routines and personal care.
- **Ambient Assisted Living (AAL)**: Uses sensors and intelligent systems to support people in their living environment.
- **Multimodal AI**: Combines information from multiple sensing modalities to understand activities and user state.
- **Personalized Assistive Systems**: Adapt support according to an individual's routines, abilities and responses.
- A continuously updated virtual representation of the individual and their behavioural state in par with the timely readings sent from sensors.

One-Line transition to Third Eye

These domains provide the foundation for Third Eye: combining multimodal sensing, personalized behavioural understanding and adaptive assistance to support independent daily living.

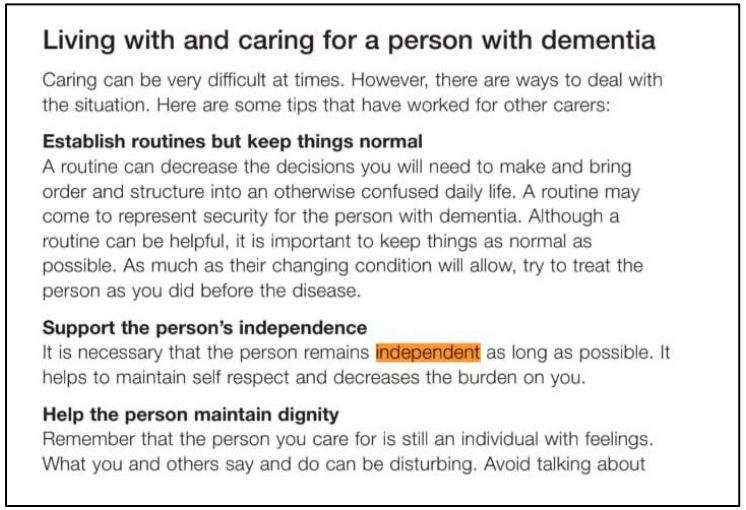
Applications/Use cases

- Independent Daily Living:
  - Supports individuals in performing routine activities independently.
  - Provides assistance only when the system identifies genuine difficulty or task interruption.
  - Reduces unnecessary dependence on caregivers for familiar tasks.
- Adaptive Dementia Support:
  - Continuously learns the person's routine, abilities, and behavioural patterns.
  - Adjusts support as the person's capabilities and needs change over time.
  - Updates the personalized model to maintain an evolving representation of the individual.
- Caregiver Support:
  - Provides caregivers with meaningful information about changes in behaviour, routines, and task difficulties.
  - Escalates situations when patient-facing assistance is insufficient.
- Long-Term Behavioural Understanding:
  - Maintains a history of routine changes, difficulties, interventions, and responses.
  - Distinguishes temporary deviations from persistent behavioural changes.
  - Supports more personalized assistance as the individual's condition evolves.



Dementia-inclusive societies

Fostering inclusive and age- and dementia-friendly environments is an important way to promote respect, acceptance, participation, safety and inclusion of people with dementia in a manner that meets their needs and those of their families and carers. This means ensuring that people with dementia and their carers can participate fully in society and are empowered to live independently and enjoy respect, freedom, dignity, equality, accessibility and quality of life. Making changes to the social and built environment – such as home modifications, opportunities for social, cultural and community participation, and the provision of person-centred health and social care – is key to this process.



Living with and caring for a person with dementia

Caring can be very difficult at times. However, there are ways to deal with the situation. Here are some tips that have worked for other carers:

Establish routines but keep things normal
A routine can decrease the decisions you will need to make and bring order and structure into an otherwise confused daily life. A routine may come to represent security for the person with dementia. Although a routine can be helpful, it is important to keep things as normal as possible. As much as their changing condition will allow, try to treat the person as you did before the disease.

Support the person's independence
It is necessary that the person remains independent as long as possible. It helps to maintain self respect and decreases the burden on you.

Help the person maintain dignity
Remember that the person you care for is still an individual with feelings. What you and others say and do can be disturbing. Avoid talking about

Image References
https://iris.who.int/server/api/core/bitstreams/8ef51bbf-75bb-4136-be49-5808a354a35b/content
https://iris.who.int/server/api/core/bitstreams/9e8aab50-ca34-45e0-9fb0-26e675746f65/content

Applications/Use cases



Maximize Independence

Alzheimer's and other dementias are associated with the gradual loss of cognition and ability to perform the functions of everyday life. As a result, many individuals feel overwhelmed by the sense of loss surrounding their autonomy. One of the greatest challenges care partners face in the early stage of Alzheimer's is not knowing when and how much assistance to provide. For individuals who live alone, changes in the ability to independently perform activities of daily living will require the support of others to ensure their safety.

PERSPECTIVES OF PEOPLE LIVING WITH DEMENTIA

People living in the early stage of Alzheimer's felt strongly about remaining independent for as long as possible. They see care providers as playing an important role in helping to maximize their independence, but want them to provide assistance "in the right ways." As one person stated, **"Let me do as much as I can [by myself] and help when I need it or get confused."**

Determining *how* care providers provide help was identified as critically important in helping those living with the disease maintain a sense of autonomy and control over their life. Individuals in the early stage reported that they need ongoing communication with care providers as Alzheimer's progresses to identify coping strategies that foster their independence. As reported by one individual, "Try different things to help me navigate the uncertainties that lie ahead." Knowing what type of assistance is needed and how the individual wants to be helped are considered important factors in providing person-centered care. For example, one individual commented, **"Ask me HOW I want you to help me,"** while another stated, **"Ask me what I am comfortable doing on my own and what activities I think I need help with."**

> **"Let me do as much as I can [by myself] and help when I need it or get confused."**

> **"Ask me what I am comfortable doing on my own and what activities I think I need help with."**

Image Reference
https://www.alz.org/getmedia/a6b80947-18cb-4daf-91e4-7f4c52d598fd/quality-care-person-living-with-dementia

Expected Deliverables

Capstone-I Deliverables

Research & System Foundation

- Literature & Domain Study: Review dementia, independent living, assistive technologies, behavioural modelling, and Digital Twin approaches.
- Requirement & Scope Definition: Identify system requirements, use cases, constraints, and feasibility.
- Data & Sensing Study: Study relevant datasets, sensing modalities, and data requirements.
- Dataset Collection: Collect datasets from reliable sources to be used for initial development.
- System Architecture: Define the overall architecture, data flow, major components, and interaction between modules.
- Initial Data/Behaviour Model: Establish the basic representation of activities, routines, behaviour, and user state.

Phase-I Outcome: A well-defined, technically feasible system design and research foundation.

Expected Deliverables

Capstone-II Deliverables

Intelligence and Adaptive Support

- Activity & Behaviour Understanding: Develop methods to identify activities, routines, deviations, and task difficulties.
- Personalized Behaviour Modelling: Build an individual baseline that can represent routine and behavioural variation.
- Context & State Understanding: Incorporate contextual and relevant physiological information for better interpretation.
- Assistance Decision Framework: Develop the logic for deciding when assistance is needed and the appropriate level of support.
- Response & Adaptation: Model user responses to assistance and use them to improve future support.

Phase-II Outcome: A functional intelligent core capable of understanding behaviour and generating personalized assistance decisions.

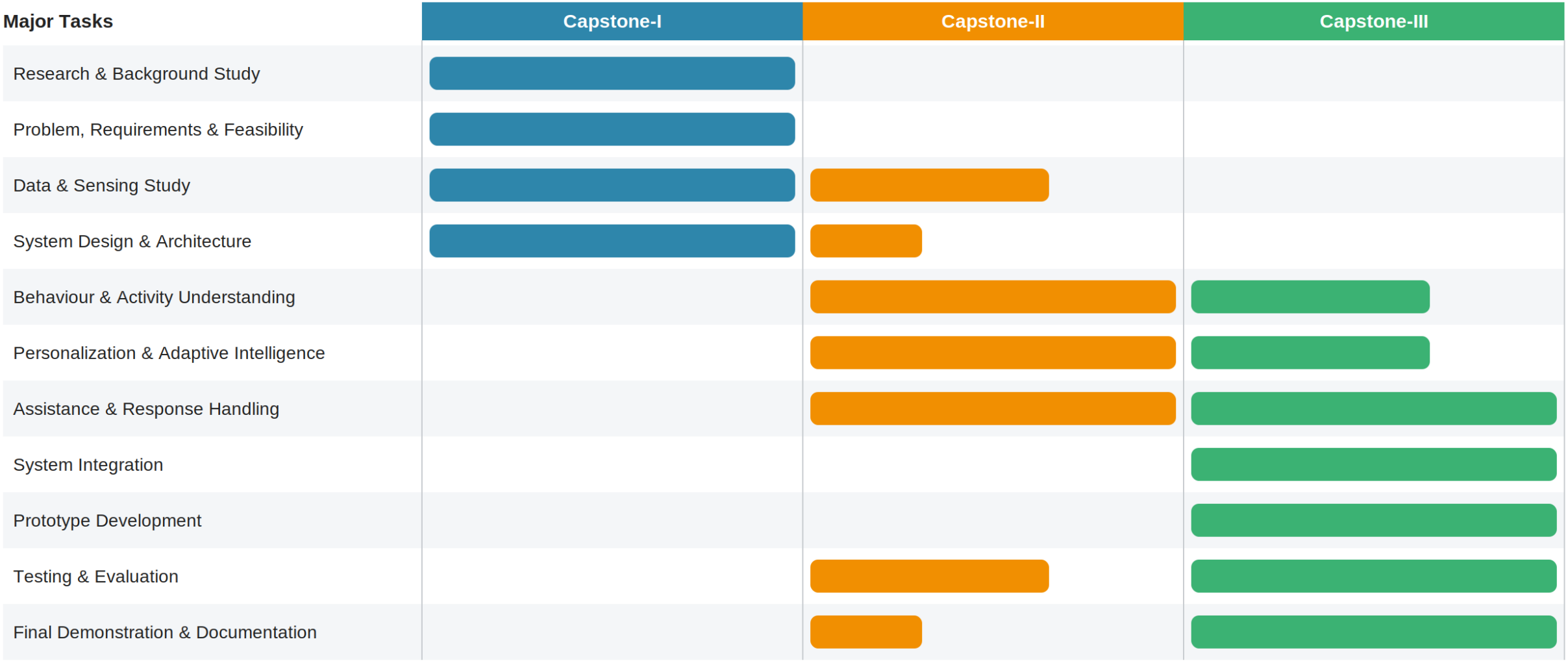
Expected Deliverables

Capstone-III Deliverables

System Integration: Integrate sensing, behavioural intelligence, assistance, learning, and user interaction modules.
End-to-End Demonstration: Demonstrate the complete **observe** → **understand** → **assist** → **learn** cycle using representative scenarios.
Prototype Implementation: Integrate feasible hardware/software components into a working prototype.
System Evaluation: Evaluate functionality, adaptability, assistance behaviour, and overall system performance.

Phase-III Outcome: An integrated prototype demonstrating the proposed personalized adaptive assistive system.

Capstone (Phase-I & Phase-II) Project Timeline



Capstone (Phase-I & Phase-II) Project Timeline

1	Sensing & Activity Understanding	Sensor/data study Data preprocessing Activity & ADL understanding Sensor feasibility & integration
2	Behaviour & Personalization	Behaviour and Personal routine modelling Behavioural changes/deviation Context understanding Personalization & adaptation
3	Assistance & User Interaction	Assistance decision logic Assistance levels and appropriate guidance Response detection and adaptation
4	System Integration & Evaluation	Overall system integration (end-to-end workflow) Prototype development Testing & evaluation Demonstration & documentation

Thank You